

Question 1

A Generative AI Engineer needs to generate responses to user queries that align with a company's branding and tone. What approach ensures the LLM aligns with these requirements? **(Choose two)**

1. Train a classification model to label responses by tone.

Explanation

Incorrect: Classification does not modify responses to align with branding.

2. Adjust the model temperature to achieve desired creativity.

Explanation

Incorrect: Temperature adjustments affect randomness but do not enforce tone or branding.

3. Use a summarization model to simplify responses.

Explanation

Incorrect: Summarization does not address tone or branding.

4. Use metaprompts to explicitly define the desired tone and style.

Explanation

Correct: Metaprompts guide the model to generate outputs consistent with the company's branding.

5. Fine-tune the LLM on company documents and communication guidelines.

Explanation

Correct: Fine-tuning ensures the model aligns with the company's branding and tone.

Overall explanation

Combining fine-tuning with metaprompts ensures the LLM generates responses aligned with the company's tone and branding.

Question 2

A Generative AI Engineer is building a RAG (Retrieval-Augmented Generation) system for a legal firm. The system retrieves relevant clauses from contracts and summarizes them for lawyers. Which component is most critical to ensure relevant clauses are retrieved?

1. Build a rule-based system for clause extraction.

Explanation

Incorrect: Rule-based systems are inflexible and cannot handle nuanced queries effectively.

2. Use a keyword-based search algorithm.

Explanation

Incorrect: Keyword searches are rigid and may miss relevant clauses that use rephrased language or synonyms.

3. Fine-tune an LLM for legal documents.

Explanation

Incorrect: Fine-tuned LLMs generate summaries but depend on effective retrieval for relevant input.

4. Implement a vector database for semantic retrieval of clauses.

Explanation

Correct: Vector databases enable semantic search, retrieving clauses based on meaning rather than exact keyword matches.

Overall explanation

Vector databases ensure retrieval of contextually relevant clauses, providing accurate inputs for generating high-quality summaries.

Question 3

A Generative AI Engineer must deploy a Retrieval-Augmented Generation (RAG) application for technical documentation. What is the first step in the deployment process?

1. Fine-tune the LLM for summarization.

Explanation

Incorrect: Fine-tuning enhances performance but does not precede document embedding.

2. Configure the endpoint for inference.

Explanation

Incorrect: Endpoint configuration is a later deployment step.

3. Create vector embeddings for the technical documents.

Explanation

Correct: Embedding the documents is the first step to enable effective similarity-based retrieval.

4. Register the model in Unity Catalog.

Explanation

Incorrect: Registration comes later and is not foundational to the retrieval pipeline.

Overall explanation

Creating embeddings is the foundation of the retrieval process, enabling RAG applications to search and retrieve relevant documents effectively.

Question 4

A Generative AI Engineer needs to design a prompt for an LLM to provide personalized fitness advice. The advice should include a recommended workout routine, calorie target, and hydration tips. What is the most effective prompt?

1. Explain general fitness principles to the user.

Explanation

Explaining general fitness principles to the user is not the same as providing personalized fitness advice that includes a workout routine, daily calorie target, and hydration tips. This prompt does not address the specific requirements for personalized recommendations and may result in generic advice that does not meet the user's individual needs.

2. Generate fitness advice based on user data.

Explanation

This prompt is too vague and does not provide clear instructions on how the fitness advice should be personalized. It lacks specificity in terms of the workout routine, calorie target, and hydration tips, which are essential components of effective personalized fitness advice.

3. Summarize the user's fitness goals into actionable steps.

Explanation

While summarizing the user's fitness goals into actionable steps is important, this prompt does not explicitly mention the need for a workout routine, daily calorie target, and hydration tips. It may not provide the LLM with enough information to generate comprehensive personalized fitness advice.

4. Provide personalized fitness advice, including a workout routine, daily calorie target, and hydration tips, based on user inputs.

Explanation

This prompt clearly outlines the specific requirements for the personalized fitness advice, including a workout routine, daily calorie target, and hydration tips based on user inputs. It ensures that the LLM understands the need for personalized recommendations tailored to individual user needs.

Overall explanation

Explicit prompts detailing required fields ensure the LLM generates personalized, comprehensive fitness advice.

Question 5

A Generative AI Engineer must design a chain for a chatbot that uses LangChain to process user inputs, retrieve relevant data, and generate conversational responses. What should the chain include?

(Choose two)

1. A prompt template for structuring user inputs, a retriever for relevant data, and an LLM for generating responses.

Explanation

Correct: These components enable structured inputs, accurate retrieval, and dynamic response generation.

2. A classifier to identify user intents for retrieval.

Explanation

Correct: Classification improves retrieval by aligning user intents with data.

3. Pretrained embeddings without retrievers.

Explanation

Incorrect: Embeddings alone cannot retrieve or process user inputs dynamically.

4. A summarization model to reduce response length.

Explanation

Incorrect: Summarization may remove critical details, reducing response quality.

5. Hardcoded templates for fixed responses.

Explanation

Incorrect: Hardcoded templates lack flexibility for diverse user queries.

Overall explanation

Combining prompt templates, retrievers, LLMs, and classifiers ensures a dynamic and context-aware chatbot chain using LangChain.

Question 6

A Generative AI Engineer is tasked with improving the quality of a RAG system by addressing inflammatory outputs in its responses. What is the most effective action to mitigate offensive outputs?

1. Restrict access to the data sources to a limited number of users.

Explanation

Incorrect: Limiting access does not directly prevent offensive outputs.

2. Increase the frequency of upstream data updates.

Explanation

Incorrect: More frequent updates may not address the quality of the data itself.

3. Inform the user of the expected RAG behaviour.

Explanation

Incorrect: Informing users sets expectations but does not mitigate offensive outputs.

4. **Curate upstream data through manual review before ingestion.**

Explanation

Correct: Curating upstream data ensures that the training and retrieval sources are free of problematic content.

Overall explanation

Properly curated data is essential for mitigating offensive outputs and improving RAG system quality.

Question 7

A Generative AI Engineer must deploy a Retrieval-Augmented Generation (RAG) application for e-commerce customer support. What are the critical steps to set up this system?

1. 1.Build a retriever → 2. Preprocess customer queries → 3. Deploy a summarizer → 4. Store embeddings in a database.

Explanation

Incorrect: Retrieval workflows require embeddings and indexing to be set up first.

Correct answer

2. **1.Generate embeddings for product descriptions → 2. Store embeddings in a Vector Search index → 3. Integrate the retriever with an LLM → 4. Deploy an endpoint for the pipeline.**

Explanation

Correct: This sequence ensures product descriptions are indexed for retrieval and combined with an LLM for dynamic customer query responses.

3. 1.Fine-tune the LLM on customer reviews → 2. Build a retriever → 3. Generate embeddings → 4. Deploy the system.

Explanation

Incorrect: Embeddings and indexing must precede retriever development for effective retrieval.

4. 1.Train a classification model for product categories → 2. Deploy a retriever → 3. Summarize queries → 4. Store outputs.

Explanation

Incorrect: Classification models are not required for RAG systems focused on retrieval and generation.

Overall explanation

The correct sequence includes embedding generation, indexing, retrieval integration, and deploying an endpoint to serve customer queries dynamically.

Question 8

A Generative AI Engineer must deploy a multi-user RAG application that analyzes scientific research articles. The application should support concurrent queries while protecting sensitive data. What measures should they implement? **(Choose two)**

1. Use public vector stores to simplify deployment.

Explanation

Incorrect: Public vector stores may compromise data privacy and security.

2. Use shared access keys for user authentication.

Explanation

Incorrect: Shared keys are less secure and do not provide individualized access control.

3. **Encrypt data at rest and in transit to protect sensitive content.**

Explanation

Correct: Encryption prevents unauthorized access to sensitive data during storage and communication.

4. Skip concurrency testing and deploy with default settings.

Explanation

Incorrect: Concurrency testing ensures reliable performance under load.

5. **Use role-based access control (RBAC) to manage user permissions.**

Explanation

Correct: RBAC ensures secure, role-specific access to application features.

Overall explanation

RBAC and encryption ensure secure and compliant multi-user access to sensitive data in RAG applications.

Question 9

A Generative AI Engineer is designing a system to generate meeting summaries. Each summary should include participants, key decisions, and next steps. What prompt structure is most effective? **(Choose two)**

1. Use a high-temperature setting to generate creative responses.

Explanation

Incorrect: High temperature may introduce irrelevant or unpredictable content.

2. **Add an example: Participants: Alice, Bob; Decisions: Approve budget; Next Steps: Schedule follow-up.**

Explanation

Correct: Examples improve clarity and guide the model toward desired output structure.

3. Specify fields in the prompt: Provide a summary with participants, key decisions, and next steps.

Explanation

Correct: A detailed prompt ensures the LLM includes all required fields in the output.

4. Focus on condensing the text into fewer words.

Explanation

Incorrect: Conciseness does not ensure inclusion of all required elements.

5. Use a generic prompt like Summarize the meeting.

Explanation

Incorrect: Generic prompts risk omitting key fields or providing incomplete summaries.

Overall explanation

Specifying fields and providing examples ensure the LLM generates structured meeting summaries aligned with user requirements.

Question 10

A Generative AI Engineer must design a prompt template for an agent that performs database queries. The template should expose available functions such as `create_record`, `update_record`, and `delete_record`. What prompt structure is most effective?

1. Instruct the model to perform actions based on its training data.

Explanation

Incorrect: Relying on training data may result in outdated or incomplete outputs.

2. Use a generic instruction like "Manipulate database records as needed."

Explanation

Incorrect: Generic instructions lack clarity and can lead to unpredictable outputs.

3. Provide a list of available functions (`create_record`, `update_record`, `delete_record`) and instructions for using them.

Explanation

Correct: Explicitly listing functions ensures the model understands the available actions and their usage.

4. Summarize the database schema and let the model infer functions.

Explanation

Incorrect: Summarizing the schema does not guarantee accurate function use.

Overall explanation

Providing a clear list of functions and instructions ensures the model correctly interacts with the database according to the requirements.

Question 11

A Generative AI Engineer is tasked with designing a multi-agent system to automate customer queries about order statuses, delivery times, and refunds. Which tools are required to expose agent functions effectively?

1. Apply embeddings to represent order details semantically.

Explanation

Incorrect: Embeddings assist retrieval but do not handle function exposure.

2. Train a rule-based system to handle each type of query.

Explanation

Incorrect: Rule-based systems lack the flexibility needed for multi-agent tasks.

3. Build a prompt template that specifies agent capabilities and provides examples for each function.

Explanation

Correct: Clear prompt templates ensure the model accurately exposes and uses agent functions.

4. Use a summarization model to condense customer queries.

Explanation

Incorrect: Summarization does not address agent capability exposure.

Overall explanation

Prompt templates provide clear instructions, ensuring the system effectively exposes and utilizes agent functions.

Question 12

A Generative AI Engineer must filter a dataset of emails for a RAG application that extracts customer complaints. Many emails contain greetings, sign-offs, and irrelevant information. What is the best preprocessing step?

1. Group emails based on their similarity score before filtering.

Explanation

Incorrect: Clustering does not address the removal of irrelevant sections like greetings or sign-offs.

2. Retain all text to preserve context.

Explanation

Incorrect: Retaining irrelevant text reduces the system's efficiency and retrieval quality.

3. Remove standard greetings and sign-offs during preprocessing.

Explanation

Correct: Removing irrelevant text ensures that the system focuses on meaningful content, such as complaints or queries.

4. Exclude emails shorter than 50 words.

Explanation

Incorrect: Filtering by length may remove valid complaints that are concise.

Overall explanation

Filtering out greetings and sign-offs ensures the dataset focuses on meaningful text, improving RAG system efficiency and response quality.

Question 13

A Generative AI Engineer has provisioned a throughput model serving endpoint for a RAG application. They currently use a microservice to log incoming requests and outgoing responses to a remote server but are looking for a native Databricks feature to perform the same task. Which feature should they use?

1. Vector Search

Explanation

Incorrect: Vector Search is used for semantic retrieval, not request/response logging.

2. Lakeview

Explanation

Incorrect: Lakeview monitors Delta Lake but does not handle inference logs.

3. Inference Tables

Explanation

Correct: Inference Tables are specifically designed to log and monitor requests and responses for deployed model endpoints.

4. DBSQL

Explanation

Incorrect: DBSQL (Databricks SQL) is for querying data but not suitable for endpoint logging.

Overall explanation

Inference Tables provide built-in support for tracking request/response logs, making them the ideal replacement for custom microservices.

Question 14

A Generative AI Engineer is tasked with creating a multilingual chatbot for e-commerce. What factor should they prioritize when selecting an LLM?

1. Multilingual performance benchmarks.

Explanation

Correct: Multilingual benchmarks ensure the model handles diverse languages effectively for a global audience.

2. Latency.

Explanation

Incorrect: While important, latency alone does not address multilingual capability.

3. Token length support.

Explanation

Incorrect: Token length is secondary for short, conversational interactions.

4. Model size.

Explanation

Incorrect: Larger models may not always perform better on multilingual tasks.

Overall explanation

Prioritizing multilingual benchmarks ensures the LLM can handle a wide range of languages effectively for global e-commerce scenarios.

Question 15

A Generative AI Engineer is tasked with deploying an LLM in a RAG application on Databricks. The team needs to monitor both system performance and cost. Which combination of tools and techniques should they use?

(Choose three)

1. Fine-tuning the LLM on Databricks.

Explanation

Incorrect: Fine-tuning enhances accuracy but does not directly monitor performance or cost.

2. Inference logging to track token usage and response latency for each query.

Explanation

Correct: Inference logging provides insights into costs and system responsiveness, helping balance performance and expenses.

3. GPU utilization monitoring to optimize hardware performance.

Explanation

Correct: Monitoring GPU usage ensures that resources are used efficiently, reducing unnecessary costs.

4. MLflow to track retrieval metrics like precision and recall.

Explanation

Correct: MLflow logs retrieval performance, enabling the team to monitor and improve relevance.

5. Increasing context length to reduce token usage.

Explanation

Incorrect: Longer context lengths may increase token usage, raising costs.

Overall explanation

Combining inference logging, MLflow, and GPU monitoring provides a comprehensive view of system performance and cost, enabling optimization.

Question 16

A Generative AI Engineer is designing a QA system for a technical support platform. The system must include safeguards to avoid generating inaccurate troubleshooting steps. What is the most effective solution?

1. Use a summarization model to condense troubleshooting guides.

Explanation

Incorrect: Summarization does not prevent inaccuracies in generated outputs.

2. Reduce the model's temperature to limit randomness.

Explanation

Incorrect: Temperature adjustments reduce variability but do not inherently improve accuracy.

3. Implement LLM guardrails, such as verification steps and retrieval-based augmentation.

Explanation

Correct: Guardrails like retrieval-based augmentation ensure the system generates accurate and verifiable troubleshooting steps.

4. Train the LLM on technical manuals and support guides.

Explanation

Incorrect: Training improves domain knowledge but does not address the risk of inaccuracies.

Overall explanation

Implementing guardrails like verification and retrieval ensures accurate, contextually relevant troubleshooting responses.

Question 17

A Generative AI Engineer must evaluate a deployed RAG application using MLflow. Which evaluation metrics are most relevant?

1. Retrieval relevance metrics like precision and recall, and response accuracy metrics like BLEU or ROUGE.

Explanation

Correct: These metrics collectively assess retrieval and generation quality, ensuring comprehensive evaluation.

2. Throughput and cost per query.

Explanation

Incorrect: These metrics evaluate operational efficiency, not application quality.

3. Training loss and validation accuracy.

Explanation

Incorrect: These metrics are relevant for model training, not post-deployment evaluation.

4. Latency and error rates.

Explanation

Incorrect: These are deployment performance metrics, not evaluation metrics for quality assessment.

Overall explanation

Precision, recall, and BLEU/ROUGE provide a holistic view of retrieval and generation quality in a RAG application.

Question 18

A Generative AI Engineer is creating a chain in LangChain to answer customer questions about technical manuals. The system should handle lengthy manuals with a 4,096-token limit. How should the engineer structure the chain?

1. Skip retrievers and directly query the entire document.

Explanation

Incorrect: Direct queries are inefficient and less accurate without a retrieval mechanism.

2. Summarize manuals into short paragraphs before querying.

Explanation

Incorrect: Summarization may lose critical details required for precise answers.

3. Use chunking to divide manuals into manageable sections, a retriever to fetch relevant chunks, and an LLM to generate answers.

Explanation

Correct: Chunking ensures token limits are respected, while retrievers and LLMs enable accurate responses.

4. Process entire manuals without chunking.

Explanation

Incorrect: Processing lengthy documents without chunking exceeds token limits, causing failures.

Overall explanation

Chunking, retrieval, and LLMs ensure token limits are respected and responses remain accurate and detailed.

Question 19

A Generative AI Engineer is tasked with creating an AI pipeline to recommend movies based on user preferences. The recommendations must consider genres, user watch history, and ratings. How should the pipeline be structured?

1. Use a rule-based system to match users with movies.

Explanation

Incorrect: Rule-based systems lack flexibility for diverse preferences.

2. Use a classification model to predict user ratings for movies.

Explanation

Incorrect: Classification does not generate personalized recommendations.

3. Use a summarization model to highlight key points from user preferences.

Explanation

Incorrect: Summarization is insufficient for dynamic recommendation tasks.

4. Use a retriever for user watch history and an LLM to generate personalized recommendations.

Explanation

Correct: Retrievers access relevant data, and LLMs dynamically generate tailored outputs.

Overall explanation

A combination of retrievers and LLMs enables dynamic, context-aware movie recommendations based on user inputs.

Question 20

A Generative AI Engineer is tasked with filtering a dataset of academic papers for a RAG system. The papers include irrelevant sections like acknowledgments and appendices. What is the best approach?

1. Use the entire paper to maximize context for retrieval.

Explanation

Incorrect: Using the entire paper risks including irrelevant content that dilutes relevance.

2. Merge all sections into a single document for consistency.

Explanation

Incorrect: Merging sections reduces clarity and retrieval efficiency.

3. Filter out irrelevant sections like acknowledgments and appendices during preprocessing.

Explanation

Correct: Removing irrelevant sections reduces noise and improves retrieval quality.

4. Retain all sections for completeness.

Explanation

Incorrect: Retaining irrelevant sections adds unnecessary noise to the retrieval process.

Overall explanation

Preprocessing academic papers by filtering irrelevant sections ensures higher-quality retrieval in RAG systems.

Question 21

A Generative AI Engineer must ensure that their deployed RAG application adheres to enterprise compliance standards for data governance. What steps are critical? **(Choose two)**

1. Use audit logs to track endpoint queries and responses.

Explanation

Correct: Audit logs provide visibility into system usage, supporting compliance and troubleshooting.

2. Skip metadata configuration for deployed models.

Explanation

Incorrect: Metadata configuration is critical for governance and tracking compliance standards.

3. Register all models to Unity Catalog for centralized governance.

Explanation

Correct: Unity Catalog enables governance and compliance tracking for enterprise applications.

4. Use public vector stores to reduce costs.

Explanation

Incorrect: Public vector stores may compromise data security and compliance.

5. Rely only on LLMs without embedding models.

Explanation

Incorrect: Embeddings and retrieval systems are essential for effective RAG applications.

Overall explanation

Registering models and using audit logs ensure governance and compliance for enterprise-grade RAG applications.

Question 22

A Generative AI Engineer needs to deploy an LLM for product recommendations. The system must balance cost efficiency and response latency. What strategy should they implement?

1. Use the largest pre-trained model available.

Explanation

Incorrect: Large models are costly and may introduce unnecessary latency.

2. Deploy multiple models to process the same query simultaneously.

Explanation

Incorrect: This increases cost and complexity without adding efficiency.

3. Use a smaller fine-tuned model for inference while caching frequent queries.

Explanation

Correct: Smaller models reduce costs and latency, while caching improves performance for repeated queries.

4. Focus solely on reducing latency by increasing compute resources.

Explanation

Incorrect: Increasing compute resources without optimizing the model is not cost-efficient.

Overall explanation

Combining fine-tuning, model size optimization, and caching ensures a cost-effective, low-latency recommendation system.

Question 23

A Generative AI Engineer is designing a RAG application that ingests legal documents from multiple public sources. What governance steps should be taken to ensure legal compliance with the data sources?

1. Skip verifying licenses for non-commercial applications.

Explanation

Incorrect: Legal compliance is critical regardless of the application's commercial status.

2. Assume all public data can be used commercially without verification.

Explanation

Incorrect: Public availability does not equate to permissibility for commercial use.

3. **Verify the licensing terms of each data source and ensure they permit commercial use.**

Explanation

Correct: Ensuring proper licensing mitigates legal risks related to unauthorized use of public data.

4. Use the data without licensing verification but cite the sources in application outputs.

Explanation

Incorrect: Citation alone does not address the legal risks of unverified data usage.

Overall explanation

Verifying licensing terms ensures the application complies with legal requirements, avoiding potential liabilities.

Question 24

A Generative AI Engineer is building a RAG system to retrieve answers from legal documents. The documents include extraneous information like case timestamps and redundant metadata. What preprocessing strategy should they implement?

1. Exclude documents with metadata entirely.

Explanation

Incorrect: Discarding all documents with metadata risks losing valuable legal content.

2. Process all documents without filtering metadata to retain completeness.

Explanation

Incorrect: Retaining all metadata reduces retrieval precision and may lead to poor-quality outputs.

3. Remove all metadata from the documents during preprocessing.

Explanation

Incorrect: While removing metadata is helpful, some metadata, like case titles or critical dates, might still be relevant.

4. **Filter out irrelevant metadata while retaining case-specific information like case names and citations.**

Explanation

Correct: Filtering ensures only relevant information is retained, improving the retrieval quality without losing context.

Overall explanation

Removing only extraneous metadata ensures the RAG system focuses on meaningful content, improving both retrieval and generated answers.

Question 25

A Generative AI Engineer must extract text from a batch of scanned legal documents in PNG format for training a RAG system. What is the best extraction tool for this task?

1. PyPDF2 for reading scanned images.

Explanation

Incorrect: PyPDF2 cannot handle image files like PNG.

2. pdfplumber for parsing text.

Explanation

Incorrect: pdfplumber is designed for PDFs, not image-based documents.

3. **pytesseract for OCR-based text extraction from images.**

Explanation

Correct: pytesseract is a robust OCR tool for extracting text from scanned images like PNG files.

4. BeautifulSoup for parsing text.

Explanation

Incorrect: BeautifulSoup is used for HTML parsing, not OCR tasks.

Overall explanation

pytesseract enables accurate text extraction from image files, making it ideal for scanned legal documents in PNG format.

Question 26

A Generative AI Engineer is building an AI tool for summarizing meeting transcripts. The tool must identify key points, action items, and decisions. What model task is best suited for this requirement?

1. Text classification

Explanation

Incorrect: Classification maps content to categories but cannot summarize meeting transcripts.

2. Named entity recognition

Explanation

Incorrect: NER identifies entities like names or dates but does not summarize content.

3. Text summarization

Explanation

Correct: Summarization condenses meeting transcripts into concise overviews, capturing key points and decisions.

4. Sentiment analysis

Explanation

Incorrect: Sentiment analysis determines tone but does not identify meeting-specific outputs like decisions or action items.

Overall explanation

Summarization is the most suitable task for condensing meeting transcripts into concise, actionable outputs like key points, decisions, and action items.

Question 27

A Generative AI Engineer must prepare text from a series of fragmented documents and write the chunked text into Delta Lake tables using Unity Catalog. What is the correct sequence of operations?

1. Write all text as a single batch into Delta Lake tables without chunking.

Explanation

Incorrect: Unstructured input leads to inefficient storage and processing.

2. Chunk the text → Define a Delta Lake schema → Batch the chunks → Write them into Delta Lake tables.

Explanation

Correct: This sequence ensures data is chunked, structured, and efficiently written into Delta Lake tables.

3. Define the schema after writing the data into Delta Lake tables.

Explanation

Incorrect: Schemas must be defined before writing data for proper structuring.

4. Use a summarization model to preprocess the text before storage.

Explanation

Incorrect: Summarization condenses text but does not address chunking or Delta Lake requirements.

Overall explanation

This workflow ensures efficient data preparation, chunking, and storage in Delta Lake tables for Unity Catalog.

Question 28

When designing a multi-turn chatbot, why is it important to track conversation state across turns?

1. To ensure context is preserved, leading to coherent and relevant responses.

Explanation

Correct: State tracking allows the chatbot to retain key information from previous turns, improving dialogue quality.

2. To prioritize speed over context.

Explanation

Incorrect: Speed should not compromise the accuracy or coherence of responses.

3. To increase randomness in responses.

Explanation

Incorrect: Randomness does not contribute to coherence or relevance.

4. To reduce response length by limiting context size.

Explanation

Incorrect: State tracking does not inherently reduce response length.

Overall explanation

Tracking state ensures the chatbot generates contextually aware responses, improving user satisfaction and system effectiveness.

Question 29

A Generative AI Engineer is deploying a RAG application for legal document analysis. The system must support multi-user access with restricted permissions. What access control measures should they implement?

1. Rely on network-level restrictions without authentication.

Explanation

Incorrect: Network restrictions alone do not provide robust access control.

2. Use a shared access key for all users.

Explanation

Incorrect: Shared keys compromise security and lack individualized access tracking.

3. Use role-based access control (RBAC) to assign permissions based on user roles.

Explanation

Correct: RBAC provides granular control over user access, ensuring secure multi-user functionality.

4. Deploy the application without access controls to simplify development.

Explanation

Incorrect: Access controls are essential for security and compliance in multi-user systems.

Overall explanation

RBAC ensures secure and controlled multi-user access, meeting the requirements for legal document analysis systems.

Question 30

A Generative AI Engineer must deploy a Generative AI application that performs real-time language translation. Which factor is most critical for selecting the LLM?

1. High token limits.

Explanation

Incorrect: Token limits are useful for long documents but are not the primary consideration for translation tasks.

2. Multilingual training data only.

Explanation

Incorrect: While helpful, training data alone does not guarantee real-time performance or accuracy.

3. Large model size.

Explanation

Incorrect: Larger models may increase latency, reducing suitability for real-time applications.

4. Low latency and high translation accuracy.

Explanation

Correct: Low latency ensures real-time responses, and high accuracy ensures translation quality.

Overall explanation

Prioritizing low latency and accuracy ensures the LLM delivers high-quality translations in real-time, suitable for this application.

Question 31

A Generative AI Engineer is building a chatbot for troubleshooting software issues. What guardrail strategy can prevent the chatbot from suggesting harmful actions?

1. Allow the chatbot to generate unrestricted responses.

Explanation

Incorrect: Unrestricted responses increase the risk of generating harmful suggestions.

2. Use predefined filters to exclude unsafe or potentially harmful responses.

Explanation

Correct: Guardrails like predefined filters help ensure the chatbot avoids generating unsafe suggestions.

3. Reduce the length of chatbot responses.

Explanation

Incorrect: Shorter responses do not inherently reduce harmful content.

4. Prioritize response speed over accuracy.

Explanation

Incorrect: Speed does not ensure safety or relevance.

Overall explanation

Predefined filters act as guardrails, preventing unsafe suggestions and ensuring the chatbot remains reliable and safe to use.

Question 32

A Generative AI Engineer is tasked with creating a pipeline to store structured text extracted from scanned receipts in Delta Lake tables. What sequence ensures consistency and efficiency?

1. Use regex on the raw text and write results directly to Delta Lake.

Explanation

Incorrect: Regex alone cannot handle all complexities of structured text like receipts.

2. Manually clean and load data into Delta Lake tables.

Explanation

Incorrect: Manual steps are time-intensive and inefficient for large-scale pipelines.

3. Define a schema, apply OCR to extract text, and write structured data to Delta Lake using Spark.

Explanation

Correct: Defining a schema ensures consistency, OCR allows text extraction, and Spark enables scalable data writing.

4. Write raw extracted text into Delta Lake without defining a schema.

Explanation

Incorrect: Writing unstructured text without a schema risks inconsistencies and poor data usability.

Overall explanation

A structured pipeline using a schema, OCR, and Spark ensures reliable, scalable, and efficient data ingestion into Delta Lake tables.

Question 33

A Generative AI Engineer is tasked with designing a multi-user application where different teams access the same LLM. How can the engineer ensure appropriate access controls?

1. Assign role-based permissions that control access to specific functionalities of the application.

Explanation

Correct: Role-based permissions ensure that each user or team can access only the parts of the application relevant to their role, enhancing security.

2. Enable all users to access the full application.

Explanation

Incorrect: Universal access increases the risk of data exposure and misuse.

3. Use encryption to restrict unauthorized actions.

Explanation

Incorrect: Encryption protects data at rest or in transit but does not manage access permissions.

4. Implement temperature settings for user-specific queries.

Explanation

Incorrect: Temperature settings affect randomness, not access control.

Overall explanation

Role-based permissions provide robust access controls, ensuring that users interact with only the parts of the application appropriate to their needs.

Question 34

A Generative AI Engineer is tasked with coding a simple chain using LangChain to query a customer database. The chain must extract user details based on their email ID. What components are essential?

1. A rule-based system for hardcoded email queries.

Explanation

Incorrect: Rule-based systems lack flexibility for dynamic inputs.

2. A retriever to query the database, an LLM to process responses, and a prompt template for query generation.

Explanation

Correct: These components ensure effective querying and response generation based on user inputs.

3. A classification model to categorize user emails.

Explanation

Incorrect: Categorization does not help retrieve user details from a database.

4. A summarization model to condense query results.

Explanation

Incorrect: Summarization is unnecessary for extracting specific details like user information.

Overall explanation

Combining retrievers, LLMs, and prompt templates ensures dynamic querying and accurate response generation using LangChain.

Question 35

A Generative AI Engineer needs to monitor the performance of a deployed LLM in production. What key metrics should they prioritize?

1. BLEU and perplexity.

Explanation

Incorrect: These metrics focus on language quality and training performance, not operational monitoring.

2. Latency, throughput, and error rate.

Explanation

Correct: These metrics ensure the system's responsiveness, capacity, and reliability in production.

3. Token usage and input-output length ratios.

Explanation

Incorrect: While useful for cost analysis, these metrics do not monitor deployment performance directly.

4. Accuracy and F1 score.

Explanation

Incorrect: These are task-specific evaluation metrics, not general deployment monitoring metrics.

Overall explanation

Latency, throughput, and error rate provide critical insights into the operational performance of a deployed LLM.

Question 36

What is the primary benefit of registering a model's signature when deploying an endpoint?

1. It increases model training efficiency.

Explanation

Incorrect: Model signatures are unrelated to training processes.

2. It reduces latency during inference.

Explanation

Incorrect: Signatures ensure compatibility but do not impact latency directly.

3. It ensures that the model receives inputs and returns outputs in the correct format.

Explanation

Correct: Model signatures define the expected input and output structure, ensuring compatibility during inference.

4. It enables real-time logging of model performance.

Explanation

Incorrect: Logging is separate from the function of model signatures.

Overall explanation

Model signatures define input and output formats, ensuring that the model operates correctly during deployment and inference.

Question 37

A Generative AI Engineer is tasked with designing a pipeline to generate personalized travel itineraries based on user preferences. The system must consider budget, preferred destinations, and activity types. What is the best approach? **(Choose two)**

1. Use an LLM to generate complete itineraries tailored to user inputs.

Explanation

Correct: LLMs dynamically create personalized travel plans based on contextual information.

2. Use a rule-based system to match users with predefined itineraries.

Explanation

Incorrect: Rule-based systems lack the flexibility needed for personalization.

3. Use an embedding model to represent user preferences and destinations contextually.

Explanation

Correct: Embeddings capture the relationship between user preferences and available options.

4. Use a classification model to categorize user preferences by travel type.

Explanation

Incorrect: Categorization does not generate full itineraries.

5. Use a summarization model to condense user inputs into concise descriptions.

Explanation

Incorrect: Summarization does not create detailed plans.

Overall explanation

Combining embeddings and LLMs ensures dynamic and contextually relevant travel itineraries tailored to user needs.

Question 38

A Generative AI Engineer needs to assess the quality of an LLM's output for generating product descriptions. What qualitative assessment should they focus on?

1. Response generation speed.

Explanation

Incorrect: Speed is secondary to quality in product description tasks.

2. Number of unique words used.

Explanation

Incorrect: Word variety does not ensure quality or relevance.

3. Length and style consistency.

Explanation

Incorrect: While helpful, length and style alone do not guarantee quality.

4. Relevance, creativity, and factual accuracy.

Explanation

Correct: These factors ensure the generated product descriptions are accurate, engaging, and aligned with the input data.

Overall explanation

Evaluating relevance, creativity, and factual accuracy ensures high-quality product descriptions that meet user expectations.

Question 39

A Generative AI Engineer must create metaprompts for a language model to prevent generating toxic content in a social media moderation system. What should the metaprompts emphasize?

1. Response diversity to improve engagement.

Explanation

Incorrect: Diversity is unrelated to toxicity prevention.

2. Contextual filtering to block any toxic or harmful phrases.

Explanation

Correct: Metaprompts should guide the model to recognize and filter out toxic language.

3. Shorter prompts to minimize complexity.

Explanation

Incorrect: Shorter prompts do not inherently prevent toxic outputs.

4. Random sampling to reduce repetition.

Explanation

Incorrect: Randomness increases variability and may inadvertently introduce toxic content.

Overall explanation

Metaprompts emphasizing contextual filtering ensure the model avoids generating harmful or toxic content.

Question 40

A team is deploying a RAG application to assist in legal document review. The application frequently retrieves irrelevant documents despite using embeddings. What adjustment should the engineer prioritize to improve document retrieval accuracy?

1. Use a retriever optimized for semantic similarity and train the embedding model on domain-specific data.

Explanation

Correct: An optimized retriever and domain-specific embeddings align document retrieval with the specialized language and structure of legal texts.

2. Add pre-trained classification models for better retrieval.

Explanation

Incorrect: Classification models do not improve semantic similarity in retrieval tasks.

3. Increase the chunk size of documents to include more context.

Explanation

Incorrect: Larger chunks may dilute relevance by including unrelated sections.

4. Implement random sampling to generate diverse retrieval results.

Explanation

Incorrect: Random sampling increases the likelihood of retrieving irrelevant documents.

Overall explanation

Domain-specific embeddings and an optimized retriever improve retrieval accuracy, ensuring relevant legal documents are surfaced efficiently.

Question 41

A Generative AI Engineer is building a pipeline for multi-turn dialogue generation in a customer support chatbot. Which approach minimizes errors over multiple turns?

1. Reset the context after every turn.

Explanation

Incorrect: Resetting context leads to disjointed conversations and user frustration.

2. Use memory or state tracking to retain user context across turns.

Explanation

Correct: Memory ensures the chatbot retains relevant context, reducing errors and improving coherence.

3. Use random sampling to increase diversity in responses.

Explanation

Incorrect: Random sampling risks incoherence and reduces conversational consistency.

4. Limit dialogue length to a fixed number of turns.

Explanation

Incorrect: Limiting dialogue length may prematurely end relevant conversations.

Overall explanation

State tracking ensures the chatbot maintains continuity and coherence throughout multi-turn interactions.

Question 42

A Generative AI Engineer is evaluating the cost performance of a Databricks-deployed LLM system for a question-answering application. The team needs to ensure scalability for increased traffic. What are the most effective strategies to monitor and optimize costs? **(Choose three)**

1. **Monitor token usage per query to identify cost-heavy operations.**

Explanation

Correct: Token usage tracking identifies areas where costs can be optimized, such as by reducing unnecessary tokens.

2. Fine-tune the LLM to handle higher throughput.

Explanation

Incorrect: Fine-tuning enhances model performance but does not directly address scalability costs.

3. **Use inference logging to analyze trends in cost and performance under high traffic.**

Explanation

Correct: Inference logging provides a detailed view of system behavior, enabling cost and performance adjustments.

4. Increase retrieval batch sizes to improve precision.

Explanation

Incorrect: Larger batch sizes do not directly optimize cost and may reduce relevance.

5. **Scale GPU resources dynamically based on demand.**

Explanation

Correct: Dynamic scaling ensures resource allocation matches traffic, avoiding unnecessary costs during low demand.

Overall explanation

Monitoring token usage, inference logging, and dynamic GPU scaling ensures cost-effective scalability for high-traffic applications.

Question 43

A retrieval-based AI system retrieves too many irrelevant documents. What preprocessing step should an engineer prioritize?

1. Randomly remove some documents to reduce system load.

Explanation

Incorrect: Random removal risks discarding useful information.

2. Increase the document chunk size to include more context.

Explanation

Incorrect: Larger chunks may introduce noise and reduce relevance.

3. **Filter documents based on relevance metrics or keywords before passing them to the LLM.**

Explanation

Correct: Filtering ensures the system processes only high-quality, relevant content, improving response accuracy.

4. Focus on documents with fewer tokens.

Explanation

Incorrect: Document length does not guarantee relevance.

Overall explanation

Filtering irrelevant documents ensures the retrieval system focuses on high-quality inputs, improving downstream AI system performance.

Question 44

A Generative AI Engineer is designing an application to generate customer reviews based on feedback forms. Some inputs contain sensitive data. How should the engineer manage this?

1. Allow sensitive inputs but ensure the model avoids generating sensitive outputs.

Explanation

Incorrect: Models cannot reliably exclude sensitive data without explicit preprocessing.

2. Use a summarization model to generalize sensitive details.

Explanation

Incorrect: Summarization does not guarantee the removal of sensitive data.

3. Process the feedback directly without masking and monitor the outputs.

Explanation

Incorrect: Processing unmasked data risks exposing sensitive information in outputs.

4. **Mask sensitive data like names and contact information in the input before processing.**

Explanation

Correct: Masking ensures sensitive data is excluded, maintaining privacy and compliance.

Overall explanation

Masking sensitive data at the input stage ensures privacy and compliance with data protection standards.

Question 45

A Generative AI Engineer is building a travel chatbot. The chatbot must answer user queries about destinations, suggest activities, and recommend accommodations. What is the best model task for this use case?

1. Summarization

Explanation

Incorrect: Summarization condenses information but does not address user queries holistically.

2. Retrieval-augmented generation (RAG)

Explanation

Correct: RAG combines document retrieval with generation, ensuring responses are accurate and based on relevant information.

3. Named entity recognition

Explanation

Incorrect: NER identifies entities but cannot answer queries or provide recommendations.

4. Text classification

Explanation

Incorrect: Classification categorizes inputs but does not generate responses or retrieve relevant documents.

Overall explanation

Retrieval-augmented generation (RAG) ensures chatbot responses are grounded in retrieved knowledge, enabling accurate and personalized recommendations for users.